🏭 Root Cause Diagnostics: Weld Lines & Fender Fitment

Executive Summary

During my engagement at **Maruti Suzuki India Limited (MSIL)** within the Manufacturing & WDE Domain, I conducted a structured root cause analysis to resolve fitment deviations affecting vehicle gap and flushness. This case study highlights the application of evidence-based analytical reasoning to complex physical systems.

1. Problem Statement

Dimensional variations in fender fitment were causing inconsistent gap and flushness during pilot and mass production phases. Surface-level observations failed to identify the underlying cause.

2. Methodology

- **Process Auditing:** Studied the complete weld line processes across underbody, side body, and framing stations.
- **Datum Point Analysis:** Traced panel datum points against Standard Operating Procedures (SOPs).
- **Feedback Loop:** Integrated trial feedback to isolate the exact station introducing the variance.

3. Key Findings (Implicit Causes)

Instead of a singular machinery failure, the RCA revealed a **sequence logic failure**. The order in which specific spot welds were applied created micro-tensions in the sheet metal, leading to cumulative dimensional deviation by the time the fender reached the fitment station.

4. Translation to AI Research

The analytical rigor required to trace a physical defect through dozens of interconnected robotic stations is directly translatable to **AI prompt debugging and reasoning trace analysis.** Finding the "implicit cause" in a hallucinating LLM requires the same structured, evidence-based decomposition that I applied at MSIL.